Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

4048292606

BIOLOGY **5090/31**

Paper 3 Practical Test **October/November 2018**

1 hour 15 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.

Electronic calculators may be used.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **9** printed pages and **3** blank pages.

© UCLES 2018

CAMBRIDGE International Examinations

In order to plan the best use of your time, read through all the questions on this paper carefully before starting work.

1 The banana is the fruit of a banana plant. As the fruit gets older, it ripens and becomes easy to eat. You are provided with a piece of a ripe banana.

- Cut a 1 cm wide cross-section from the middle to give a circular piece of banana.

(a) (i) Make a large drawing of one of the cut surfaces of the banana.

[3]

(ii) Draw a straight line across the largest diameter of your drawing. Measure this line and record the measurement.

..... mm

Measure the largest diameter on the cut surface of your banana and record the measurement.

..... mm

Calculate the magnification of your drawing. Show your working.

magnification × .....

[4]

© UCLES 2018

- Cut another thin slice from the banana to expose a fresh cross-section.
- Lay this banana section flat, with the freshly cut surface facing upwards.
- Completely cover the cut surface with iodine solution and leave this for five minutes.
- After five minutes, pick up the banana section with forceps. Use water to carefully rinse excess iodine solution into the empty container labelled **waste water**.

(b) (i) Describe your observations of the result of this test.

.....

.....

.....

.....[3]

(ii) State what you can conclude from your observations.

.....

.....

.....[2]

(c) (i) Describe how you would test the inner part of another slice of banana to see if it contains reducing sugar.

.....

.....

.....

.....

.....[3]

(ii) Carry out this test on a fresh sample of the **inner** part of the banana. If you require hot water for a water-bath, raise your hand to request it when needed. **Caution: water will be hot.**

State the result of the test and the conclusion you can make.

.....

.....

.....[2]

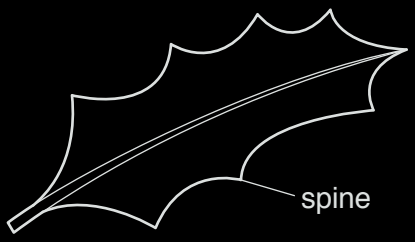
[Total: 17]

© UCLES 2018

BLANK PAGE

© UCLES 2018

2 The European holly is a tree with leaves that can have spines on their edges. The diagram shows a holly leaf with eight spines.



Some students thought that the leaves on the lower branches of a holly tree had more spines than the leaves higher up the tree. They carried out an investigation into the number of spines on the leaves at different heights. They collected leaves from three different heights of the tree, 1m, 2m and 3m above the ground, and counted the number of spines on each leaf.

Some of the data they collected is shown in the table on page 6.

© UCLES 2018

[Turn over

key
I = 1 leaf, II = 2 leaves etc.

number of spines per leaf	height above ground level/m		
	1	2	3
16	I		
15	III		
14	III		
13	III	I	
12	II		
11	II	I	
10	I		
9		I	
8		I	I
7			
6		I	II
5			I
4			II
3		II	I
2		III	IIII
1			IIII
number of leaves counted	15	.....	15
total number of spines	198	.....	48
mean number of spines per leaf	13.2	.....	3.2

© UCLES 2018

Data for five leaves collected at a height of 2m has not yet been added to the table. The five leaves are shown in the diagram.

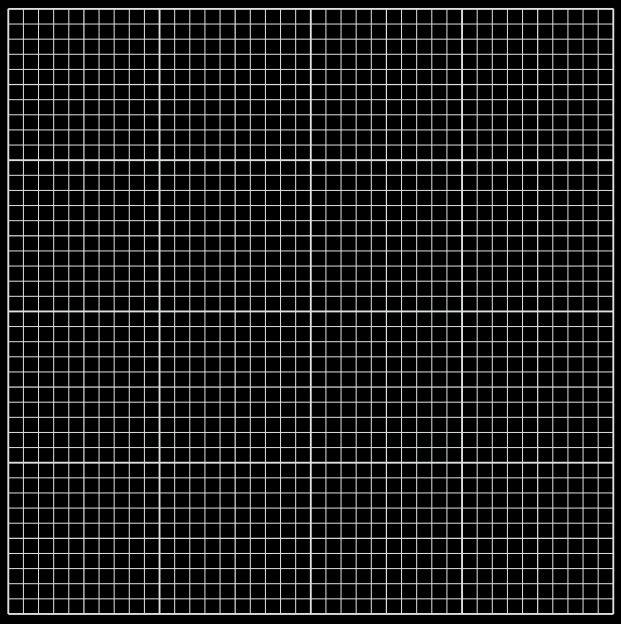


(a) (i) Count the spines on the five leaves in the diagram and enter the data in the table on page 6.

[2]

(ii) Complete the table by calculating the number of leaves counted, the total number of spines and the mean number of spines per leaf for leaves at the height of 2m. [3]

(iii) Construct a bar chart of the mean number of spines per leaf at each height, on the grid below.



[4]

(iv) State what the students could conclude from the bar chart about the effect of height on the mean number of spines.

.....

.....[1]

(v) Suggest **two** ways in which the students could have improved their investigation to give them more confidence that their conclusion was valid and reliable.

1 .....

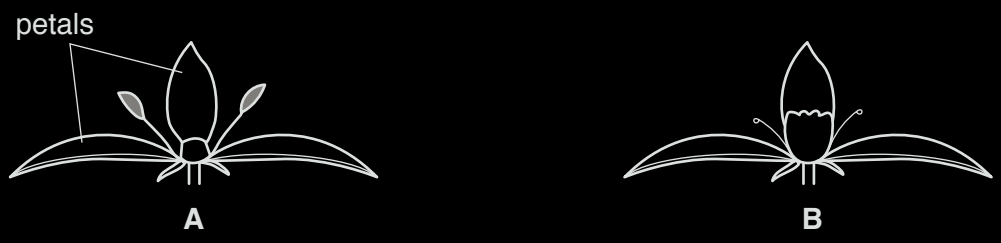
.....

2 .....

.....

[2]

(b) The holly tree is a flowering plant. It produces two types of flowers – one in which only the male parts function and one in which only the female parts function. These flowers are shown in the diagram.



magnification ×10

Complete the table to compare the stamens and carpels of these two types of flower.

flower structure	flower A	flower B
stamens		short filaments with small anthers
carpels		

[3]

[Total: 15]

3 Caffeine is a chemical present in coffee. Some people think that drinking caffeine in coffee may lead to an increase in the rate at which the heart beats.

(a) Describe how you would measure the rate at which a person's heart beats.

.....

.....

.....[2]

(b) Describe an investigation you could do to find out whether caffeine in coffee affects the rate at which the heart beats.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

(c) A student suggests that another drink contains even more caffeine than coffee. You want to design an investigation to test whether this is true. State **two** factors which you should control so that the results can be compared with the results of your coffee investigation in **(b)**.

1 .....

.....

2 .....

.....
[2]

[Total: 8]

© UCLES 2018

BLANK PAGE

© UCLES 2018

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

© UCLES 2018